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# =====
# Dataset 2: AirPassengers Time Series Analysis
# =====

# 1. Load Dataset
data(AirPassengers)
print(AirPassengers)

# =====
# 2. Convert to Data Frame
# =====
months      <- time(AirPassengers)
passengers  <- as.numeric(AirPassengers)
air_df      <- data.frame(months, passengers)
head(air_df)

# =====
# 3. Traffic Classification
# =====
traffic <- c()

for (i in passengers) {
  if (i > 400) {
    traffic <- c(traffic, "High")
  } else if (i > 250) {
    traffic <- c(traffic, "Medium")
  } else {
    traffic <- c(traffic, "Low")
  }
}

air_df$traffic <- factor(traffic)
head(air_df)

# =====
# 4. Matrix Reshape (Yearly Data)
# =====
pass_matrix <- matrix(passengers[1:120], nrow = 12)
rownames(pass_matrix) <- month.abb
colnames(pass_matrix) <- 1949:1958
print(pass_matrix)

# =====
# 5. List Structure

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# =====
air_list <- list(
  raw_data = AirPassengers,
  dataframe = air_df,
  matrix = pass_matrix
)
print(air_list)

# =====
# 6. Visualization
# =====

# Line plot (trend over time)
plot(AirPassengers,
     col = "blue",
     lwd = 2,
     main = "Air Passenger Trend")

# Histogram
hist(passengers,
     col = "purple",
     main = "Passenger Distribution")

# Boxplot
boxplot(passengers,
     main = "Passenger Spread",
     col = "orange")

```